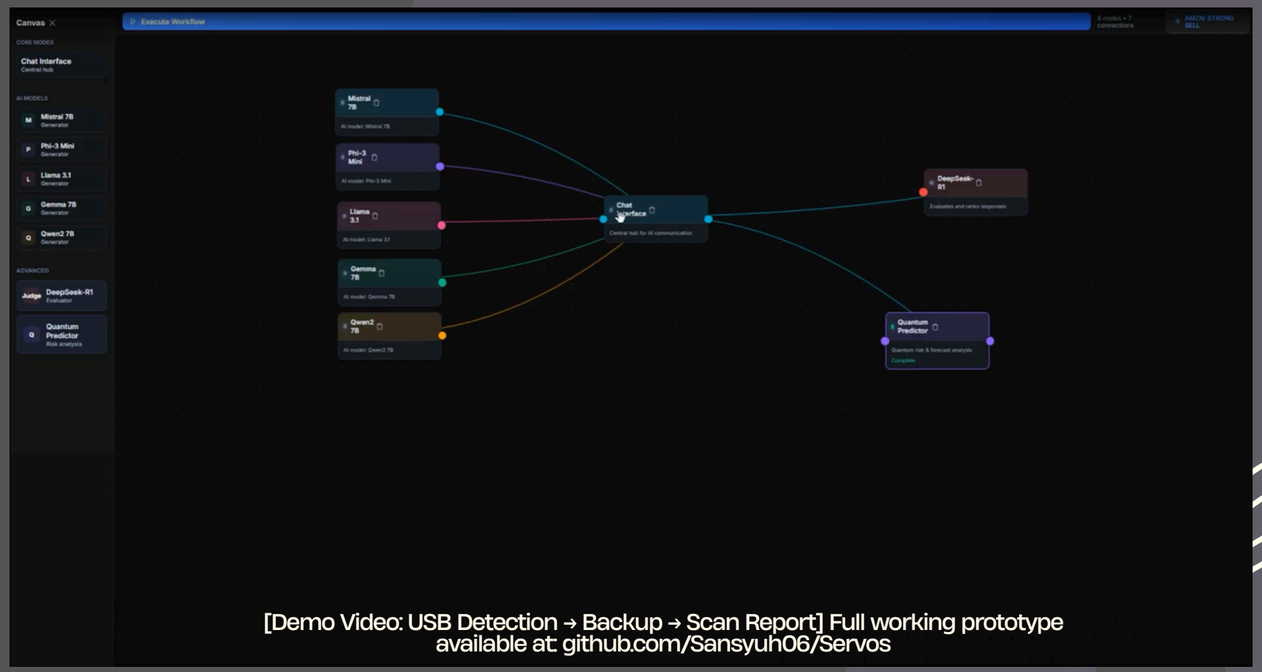
CyberHack V4

Servos

"Forensics for the Rest of Us"

Presented By: **MoMoSapiens**

Servos — Offline AI Forensic Assistant



[Demo Video: USB Detection → Backup → Scan Report] Full working prototype available at: github.com/Sansyuh06/Servos

Servos : Offline AI Forensic Assistant

Servos is an AI-powered cyber-investigation assistant that runs completely offline on the investigator's machine. It detects suspicious storage devices, enforces safe backups, and guides end-to-end forensic workflows without sending any data to the cloud designed for real-world use by Greater Chennai Police–style teams and small organizations.

Servos

The Problem – Forensics Is Slow, Risky, and Cloud Dependent

Key Challenges Investigators Face

- Sensitive evidence cannot be uploaded to online AI tools
- Many organizations lack forensic experts and expensive tools
- Manual investigation is slow and error-prone
- Evidence often lost or damaged due to panic and mistakes

Servos

Full, Hybrid, or Manual Choose Your Control Level

Full Automation
Detects events, forces backup, runs checks automatically. For beginners and rapid triage.

Hybrid Mode
Proposes each step and asks for approval. Gives investigators granular control.

Manual Mode
Provides checklists and commands. Experts execute everything themselves.

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The Suspicious USB Step-by-Step Flow

Complete Offline Forensic Workflow

1. Detect → 2. Protect (backup) → 3. Investigate (scan) → 4. Adapt (ask clarifying questions) → 5. Report (generate summary)

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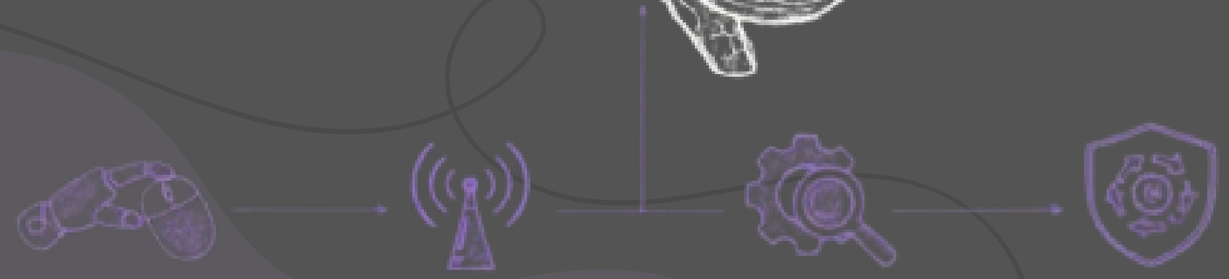
Under the Hood – How Servos Works



Pyautogui Automation | Port Monitoring | Port Detection | Agent Orchestration

Key Technology Stack

- LLM Engines: 1. Llama 2. mistral 3. phi 4. qwen 5. gemma 6. deepseek r1
- Automation Layer: pyautogui for GUI control & forced backup enforcement
- Device Detection: psutil + udev monitoring for USB/storage events
- Forensic Modules: File enumeration, hashing (MD5/SHA-256), log extraction
- Orchestration: Orca-based agent controller for workflow decisions

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Real-World Impact Serving CyberHack Mission

Air-Gapped Safety

Works offline—no cloud, no leaks, perfect for sensitive environments and regulations.

Accessibility & Guidance

Non-experts get guided workflows. Experts get fast automation. One tool, flexible control.

Evidence Preservation

Forced backups and automated procedures protect chain-of-custody and improve legal defensibility.

Servos

Status & Next Steps Serving CyberHack V4

Built for Real-World Deployment

IMPLEMENTED (v1.0):

- USB/external drive detection & monitoring
- Mandatory disk image/backup before analysis
- Guided malware & artifact scanning
- Auto-generated investigation summaries

IN PROGRESS (v1.5):

- Memory forensics (RAM dump analysis)
- Network traffic basic analysis
- Policy-based playbooks (police/enterprise/lab)

ROADMAP (v2.0):

- Offline voice interface (hands-free operation)
- Integration with Volatility 3 & Autopsy
- Multi-language support
- Law enforcement certification ready

Servos

Forensics for the Rest of Us CyberHawk Serves All

Why Servos Matters:

- No internet = no data leaks (GDPR/compliance safe)
- Guided = non-experts become competent (democratizes forensics)
- Automated = faster response, less evidence tampering
- Real-world ready = tested architecture from Multi model Orchestration

Thank you

Servos

"Forensics for the Rest of Us"

GitHub Repository: github.com/Sansyuh06/Servos

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